This assignment will not be graded and is only for practice.

Directional Derivatives:

Level 1:

1) Which of the following statements is(are) true? ***1 point***

- [ ] The directional derivative of $f(x, y)$ at a point (a, b) in the direction of a unit vector u is a non negative real number.
- [ ] The directional derivative of any function is always a scalar.
- [ ] The rate of change of $f(x_1, x_2, ..., x_n)$ in the direction of a unit vector $u = (a_1, a_2, ..., a_n)$ is called the directional derivative of f in the direction of u.
- [ ] The directional derivative of any function is a vector because it is dependent on the direction of a unit vector.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The directional derivative of any function is always a scalar.
The rate of change of $f(x_1, x_2, ..., x_n)$ in the direction of a unit vector $u = (a_1, a_2, ..., a_n)$ is called the directional derivative of f in the direction of u.

2) If the directional derivative of $f(x, y) = x + xy - 2y^2$ at the point (2,3) in the direction of the vector $(2\sqrt{2}, 2\sqrt{2})$ is $a\sqrt{2}$ where $a \in \mathrm{R}$, then find the value of a.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -3

1 point

3) If the directional derivative of $f(x, y, z) = xe^y + ye^z + ze^x$ at point (1,0,1) in the direction of a vector (2,-2,1) is $k.e$ where $k \in \mathrm{R}$, then find the value of $6k$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

4) Suppose f is a multivariable function defined on domain R^2. Which of the following is equal to the rate of change of f at point (x_1, y_1) in the direction of a vector $u = (u_1, u_2)$? ***1 point***

- $\lim_{h \to 0} \frac{f((x_1,y_1)+h(u_1,u_2))+f(x_1,y_1)}{h}$
- $\lim_{h \to 0} \frac{f((x_1,y_1)+h(u_1,u_2))-f(x_1,y_1)}{h}$
- $\lim_{h \to 0} \frac{f((x_1,y_1)+h(\frac{u_1}{\sqrt{{u_1}^2+{u_2}^2}},\frac{u_2}{\sqrt{{u_1}^2+{u_2}^2}}))-f(x_1,y_1)}{h}$
- $\lim_{h \to 0} \frac{f((x_1,y_1)+h(\frac{u_1}{\sqrt{{u_1}^2+{u_2}^2}},\frac{u_2}{\sqrt{{u_1}^2+{u_2}^2}}))+f(x_1,y_1)}{h}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{h \to 0} \frac{f((x_1,y_1)+h(\frac{u_1}{\sqrt{{u_1}^2+{u_2}^2}},\frac{u_2}{\sqrt{{u_1}^2+{u_2}^2}}))-f(x_1,y_1)}{h}$

5) If the directional derivative of $f(x, y) = xsin(y)$ at the point (1,0) in the direction of a unit vector (u_1, u_2) $(u_1, u_2 \in \mathrm{R})$ is 0.8, then find the absolute value of u_1. (Enter your answer upto one decimal place only)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Range) 0.5,0.7

1 point

Level 2:

6) Consider the function $f : \mathrm{R}^2 \longrightarrow \mathrm{R}$ defined as: $f(x,y) = \begin{cases} \frac{x^2y}{x^4+y^2} & x,y = 0 \\ 0 & x = y = 0 \end{cases}$ Find the directional derivative of f at (0,0) in the direction of the vector $u = (\frac{\sqrt{3}}{2}, \frac{1}{2})$. (Enter your answer upto one decimal place only)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Range) 1.4,1.6

1 point

7) Suppose d is the directional derivative of a scalar valued multivariable function $f(x,y) = x^2 + y^2$ at a point $P\,(1,2)$ in the direction of the line PQ, where point Q is at (4,6). The value of d is (Enter your answer upto one decimal place only)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Range) 4.3,4.5

1 point

8) Suppose $f(x,y,z) = ax^2y + yz^2 - z^2x$, where $a \in \mathrm{R}$, is a scalar valued multivariable function defined on domain $D \subset \mathrm{R}^3$. For what value of a, does the directional derivative of the given function at the point (1,1,1) in the direction of the vector $u = (1,2,-2)$ equal 3?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

9) The temperature at a point (x,y,z) is given by

$$T(x,y,z) = 10e^{-x^2+y^2-4z^2}$$

where T is measured in $^\circ C$ and x,y,z are measured in meters. If the rate of change of temperature at (2,-1,0) in the direction of a vector $u = (1,-1,1)$ is $k.\frac{e^{-3}}{\sqrt{3}}$ where $k \in \mathbb{R}$,

then the value of k is (approximate to two decimal places)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -20

1 point

10) Let $f : \mathrm{R}^2 \rightarrow \mathrm{R}$ be given by $f(x, y) = x^2 + y^2 - 2xy$. Suppose $D_u f(x_0, y_0)$ is the directional derivative at (x_0, y_0) in the direction of the unit vector $u = (\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}})$ and $D_v f(x_0, y_0)$ is the directional derivative at (x_0, y_0) in the direction of the unit vector $v = (\frac{1}{2}, \frac{\sqrt{3}}{2})$. Consider a matrix $A = \begin{bmatrix} D_u f(1,2) & D_u f(2,1) \\ D_v f(1,2) & D_v f(2,1) \end{bmatrix}$. What is the value of $det(A)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Your score is: 0/10